CSC488 - Captsone Project

1 CSC 488: Captsone Project

- Union College, Fall 2026
- Georgia Doing, PhD
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- office: Steinmetz 217

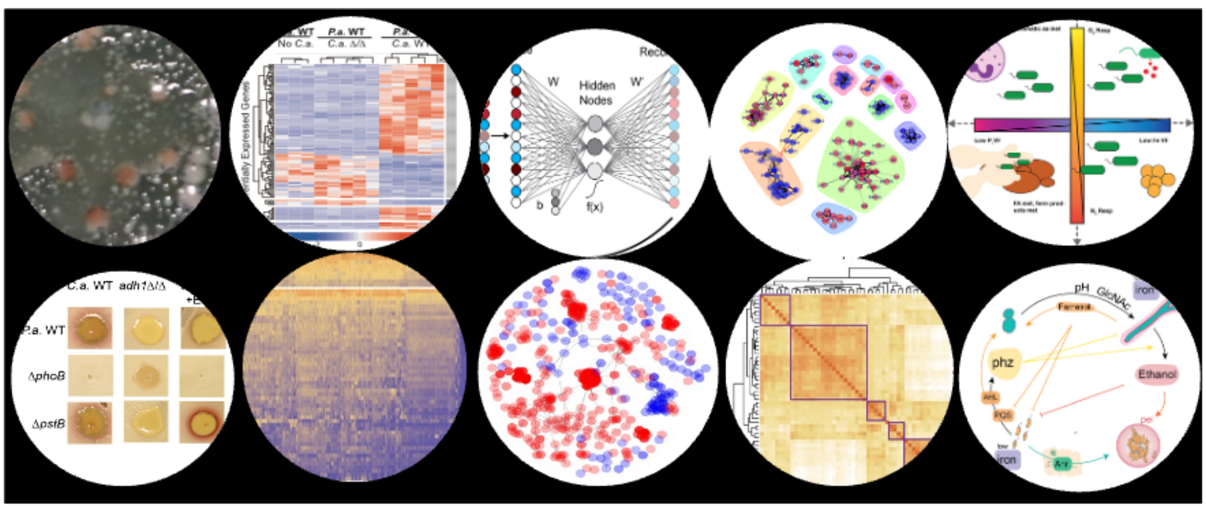

Figure 1: Views of Biological Data.

2 COURSE BASICS

- Lab Meeting: Thursday 4:30 am - 5:30 pm
  - Location: ISEC 070
- Research Office Hours:
  - Location: ISEC 070
  - Tuesday 2:00 pm - 4:00 pm
- Office Hours:
  - Location: Steinmetz 217
  - Monday 4:00 pm - 5:30 pm
  - Thursday 3:00 pm - 3:30 pm
  - subject to change, check course website for most up-to-date schedule
  - drop-in or schedule a 15 minute slot: https://calendar.app.google/8bus6pfDvyphR9ar5
  - by appointment for another time or over zoom

3 OVERVIEW

In CSC-488, you will complete a capstone project that applies your CS knowledge.

3.0.1 Section Topic: Assessing Explainability in Biological Data Mining – Georgia Doing

Mining biological data has been aided by machine learning models that are particularly good at picking up non-linear relationships, mirroring molecular mechanisms like positive and negative feedback loops and threshold effects. We will use post-hoc explainability algorithms to trace mined patterns and determine which result from artifice and which are testable, with the potential to uncover novel biology.

Individual projects with no prerequisites

4 LEARNING OUTCOMES

After successfully completing this course, you will be able to:

1. design and carry out a CS project from problem formulation to solution and evaluation.
   a. formulate and refine research questions related to the section topic and apply appropriate CS methods to address them;

   b. apply theoretical and practical techniques to develop in-depth solutions within a CS subfield; and
   c. analyze and evaluate the effectiveness of your solutions to such problems.

2. communicate technical concepts effectively to CS audiences in both written and oral forms.

   a. read and interpret publications in the relevant sub-field of CS;
   b. present a clear and coherent narrative of a project, including its motivation, methods, and outcomes;
   c. tailor communication style and content to the knowledge level and interests of the intended audience (senior computer science majors and faculty); and
   d. use LaTeX to produce technical documents that meet CS academic standards.

5 COURSEWORK

The course grade is split into two parts, process and products, each part being further broken down into subcategories. You will be graded individually, even if you are working as part of a group project. Link to the rubric. You will receive feedback from your instructor and other CS faculty at various points during the term. This includes a midterm check-in using the rubric following the elevator pitches in Week 4.

5.1 Process:

This encapsulates all the work you do to learn about the subject matter and apply your knowledge to solve the problem.

- See the rubric for details

5.1.1 Project Success (25%):

Your project will be evaluated on how effectively it defines and addresses a meaningful, non-trivial problem in computer science. A successful project will demonstrate thoughtful research design, technical execution, and analysis grounded in the CS literature and methods. Specifically, your work will be assessed based on the following:

1. Problem Definition: You clearly identify and refine a research question that is feasible, relevant to your section's topic, and appropriately scoped.
2. Research Design: You develop a methodology to answer your question, drawing on techniques, tools, or frameworks from prior CS research and literature.

3. Execution and Results: You follow your proposed methodology and produce meaningful results through implementation, experimentation, analysis, or other outcomes appropriate to your project type.
4. Analysis and Evaluation: You critically interpret your results, assess their limitations, and reflect on their implications in the context of your research question and the broader field.
5. Holistic Success: As a whole, your project successfully addresses the problem and demonstrates depth and completeness.

Projects may take many forms, including software prototypes, theoretical contributions, applied systems, or analytical studies.

5.1.2 Habits for Success (25%):

This is a 400-level course that satisfies the College's and the CS department's senior writing requirement. You will demonstrate professionalism and effective work habits throughout the research process: exhibit independence, initiative, and sustained effort in pursuing research goals; manage time and responsibilities effectively; collaborate constructively with others; and reflect on personal growth, setbacks, and learning throughout the project.

Your performance will be evaluated based on how well you demonstrate the following habits for success:

- Show persistence and dedication by making steady progress and maintaining consistent effort throughout the term.
- Approach obstacles independently and creatively, seeking help and resources proactively when needed.
- Apply both technical and theoretical knowledge and skills gained from previous CS courses effectively in your project work.
- Reflect critically on your accomplishments, learning from setbacks, and identifying areas for improvement.
- Actively revise and improve your work through multiple iterations based on feedback.
- Collaborate productively and respectfully with your peers, contributing constructively to group efforts if applicable.
- Attend all scheduled CS seminars to engage fully with the course community and learning opportunities.

These expectations support your growth as a capable and responsible computer scientist ready to tackle complex problems with initiative and intellectual maturity.

5.2 Products:

This encapsulates all the work to communicate what you have done. There are two components of this: two public presentations (20%) and the final report (30%).

5.2.1 Public Presentations:

You will present your work at two main stages of the project:

1. Elevator Pitch (5%):

- When: During CS seminar slot in Week 4
- Duration: 2 minutes for student presentation followed by 2 minutes Q&A.
- Format: Each student will give a short pitch introducing your proposed project, including the research question, its significance, and your initial plan. This will be an individual presentation. This is meant to help you clarify your direction early and get initial feedback from peers and instructors.
- Grading: The elevator pitch will be evaluated based on whether you:
  - present a clear and convincing story about your research question and how you plan to answer it, and
  - organize and deliver your presentation in a way that effectively reaches your intended audience.

2. Preliminary Exam & Showcase (15%):

- When:
  - Review: 20 minutes in Week 9 at a time you schedule
  - Showcase: 60 minutes in Week 10 during CS Seminar
- Format:
  - You will be assigned a CS faculty member different from your CSC 488 instructor to review the progress of your project.
  - You will schedule a meeting time with your faculty examiner and your 488 instructor to occur in Week 9. You are responsible for scheduling this: reach out to your assigned faculty examiner by Week 8 at the latest. Preliminary exam meetings will happen individually, even for group projects, but showcase posters may be common.
  - You will prepare a showcase poster communicating the progress you have made on your project so far. It is expected that your poster draft will contain results and analysis even if they are still preliminary. There are Latex and PPT poster templates available on the CS Thesis webpage.

    - At least 24 hours in advance of your preliminary exam, you will share a complete draft of your showcase poster with your faculty examiner.
    - During the meeting you will begin by briefly sharing (2-3 minutes) your project with the examiner. You can expect that your examiner will have reviewed your poster before your meeting.
    - You and the examiner will discuss your project. You will be asked detailed questions about it. The examiner will provide you with verbal feedback on both the state of the project, but also how you've communicated this in your poster and during the exam meeting.
    - You are expected to take detailed notes on the feedback you received and incorporate it into the final version of your showcase poster.
    - In Week 10 you will publicly present your finalized poster at the CS Seminar.
    - You will be asked to hang your showcase poster in public view in CS spaces around campus, e.g., the 2nd floor Steinmetz Hallway, Crochet, CAP, Pasta, or Olin 107.
- Grading: Your instructor will use both the outcome of your preliminary exam and your final showcase poster to determine your grade for this section based on whether you:
    - communicate a coherent and convincing argument grounded in evidence,
    - clearly present your project, including its methodology, results, and what those results mean,
    - articulate the contribution your project makes to the field or problem domain,
    - organize and deliver your presentation in a way that effectively reaches your intended audience, and
    - make appropriate changes to your poster and communication in response to the feedback you received.

5.2.2 Final Report (30%):

Due Date: Last day of finals week This report will document the complete research process, including background, methodology, results, and reflection. You must use the provided LaTeX template to format your final report. Your final report will be evaluated based on how effectively it does the following:

- tells a coherent and complete story of your project
- clearly states your research question
- explains your methodology in a detailed and understandable way
- clearly presents your results
- provides thoughtful analysis and well-supported conclusions
- uses relevant literature as evidence and context
- articulates the contribution your project makes to the field or problem domain

- organizes your writing in a logical and readable way
- targets the appropriate audience (computer science readers)
- uses appropriate technical writing style
- shows evidence of having been revised and improved through multiple rounds of editing and feedback

Please note: You are required to submit a title, abstract, and the final version of your report for inclusion of completed projects on our Computer Science website.

6 Requirements for Honors in Computer Science

To earn Departmental Honors in Computer Science, you must maintain a minimum 3.3 overall GPA and a 3.3 major GPA AND fulfill these two requirements in CSC 488:

- Grade: Earn an A- or higher in CSC 488 or IDM 488 / IDM 499.
- Oral Public Presentation: Deliver an oral presentation of your capstone project. Note: It has to be an oral presentation. Poster presentations do not qualify.
  - If you are completing your capstone project in Fall or Winter: You must present at the Steinmetz Symposium in the Spring term.
  - If you are completing your capstone project in Spring: Present at a separate, department-scheduled oral presentation during Week 10 of Spring term.

7 Alice P. and Donald C. Loughry (1952) Prize in Computer Science

Each year, the Alice P. and Donald C. Loughry (1952) Prize is awarded to the student who completes the best senior project in Computer Science. Your instructor may choose to nominate a student (or students) from the section for consideration. Nominations are based on the overall quality of your work, including:

- your final written report
- the technical depth and execution of your project
- presentation at preliminary exam with examiner and the extent to which examiner feedback was effectively integrated into the final project.
- final showcase presentation and public oral presentations

Note on Group Projects and Team Awards: If a group project is nominated and selected, the prize is awarded jointly to all team members. In cases where an individual team member's contribution significantly outpaces the rest of the group, the instructor may choose to nominate that specific student rather than the entire team.

8 Honor Code Statement:

Please review the official Union College honor code found here: https://www.union.edu/academic-affairs/honor-code.

In the world of CS research, you are free to build on the work of others, assuming that you appropriately credit them with it. In the case of software created by others, the situation is more complex. The license must permit reuse or derivation of the software for our academic purposes. You are expected to check the licenses of external software and ensure that our use of it is lawful before incorporating it into your project. You are encouraged and expected to (re)use the work that others have done. There are no limitations to the degree to which you can share code or writings with others in the course or collaborate to complete team goals. At the top of every source file or LATEX file, there should be a header comment mentioning all the contributors to the content of the file. If relevant, it should also include citations to all external sources used.

Your elevator pitch, final showcase, and report should contain appropriate citations of scholarly work and an acknowledgment of the work and aid others contributed to allow you to complete the task.

8.1 Generative AI

Understanding how to live and work with digital tools and platforms – from statistical software to data visualization tools to artificial intelligence tools – is an essential skill for all students in this day in age. An important part of using these tools is knowing their limits. Therefore, the use of Generative AI tools isn't allowed under any circumstances for manuscripts or code that is submitted for publication. In general, anything you submit to me for feedback, anything I read, should be from your own mind and be written by your own hand. However, in throubleshooting, I encourage you to use all the tools available to you (and that you are familiar enough with to use efficiently and effectively) to aid your learning. This includes artificial intelligence (AI) copy writing and chatbot tools such as ChatGPT, Humata.ai, DALL-E 2, and others. If you use these tools to generate text or code, you must keep track of what was generated, even if you edit it after, and what you wrote from scratch. If you are unsure of when Generative AI tools are acceptable, please consult with the professor.

Be aware that large language models like ChatGPT reproduce the inaccuracies and biases of the data they are trained on. They can provide you with results that are racist and sexist or simply false. They are also known to "hallucinate," as they synthesize, reassemble, and present information in convincing ways while misrepresenting reality. It is your responsibility to evaluate the factual accuracy and integrity of the products these tools provide.

If you use an AI tool at any point in the development and/or creation of your work for this course – you must include appropriate citations and the acknowledgment below in your Reference list:

Name of publisher/tool producer. (year). Name of AI tool (version date) [Large language model]. You must also include a full transcript of the writing or work produced by the AI tool in an appendix to your work.

Consult the Generative AI citation resources appropriate to the class or discipline. For example: MLA https://style.mla.org/citing-generative-ai/ APA: https://apastyle.apa.org/blog/how-to-cite-chatgpt. Chicago: https://www.chicagomanualofstyle.org/qanda/data/faq/topics/Documentation/faq0422.html. [Choose the appropriate model]

9 You belong here!

The CS Department welcomes individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability; and other visible and non-visible differences.

- Our department community is a place where you will be treated with respect and all members of our community are expected to contribute to a respectful, welcoming and inclusive environment for every other member.
- We acknowledge the institutionalized racism that has prevented members of marginalized groups, including black, indigenous, and other people of color (BIPOC), from fully participating in the field of CS.
- We commit to do the utmost to promote equity, inclusion, and diversity in all aspects of our discipline.
- We welcome feedback on issues that we might discuss or ways that our community can be a more just one for all people.

10 Students with Disabilities

Union College's policy is to make reasonable accommodations for qualified individuals with disabilities. If you are a person with a disability and wish to request accommodations to complete your course requirements, please make an appointment with me or stop by during my office hours as soon as possible to discuss your request. All such discussions will remain confidential.

11 COURSE RESOURCES

- Olin 107
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

12 LIBRARY RESOURCES

The library is available to help you with your research needs! Librarians can help you develop research questions, search for and select the best sources for your projects, identify research strategies, evaluate sources, and assist you with creating citations. There are multiple ways for you to contact a librarian. For more information, please see our Ask A Librarian page: https://www.union.edu/schaffer-library/ask-a-librarian.

13 CONTACTING ME OUTSIDE OF CLASS

The best method for contacting me outside of class is to stop by my office during Office Hours or whenever my office door is open. You can also schedule a time with me if my regular office hours don't work. Please contact me via email (doingg@union.edu) and we can arrange a phone or zoom call. I respond to emails as soon as possible, but it may take up to one day to get a response during the term, and longer between terms.

14 ADDITIONAL RESOURCES

Mental Health and Campus Resources

Union College is committed to supporting and advancing the mental health and well-being of our students. During the course of their academic careers, students often experience personal challenges that contribute to barriers in learning, such as drug/alcohol problems, strained relationships, chronic worrying, persistent sadness or loss of interest in enjoyable activities, family conflict, grief and loss, domestic violence, difficulty concentrating, problems with organization, procrastination and/or lack of motivation. Students also sometimes come to college with a history of learning difficulties (e.g., any form of special education), experience difficulties succeeding in a particular subject (e.g., math, reading), or have experienced some form of trauma be it emotional or physical (e.g., head injury). These mental health concerns can lead to diminished academic performance and can interfere with daily life activities. If you or someone you know has a history of mental health concerns or if you are unsure and would like a consultation, a variety of confidential services are available. The Eppler-Wolff

Counseling Center provides free counseling and therapy services (including psychiatry) to all enrolled students. Please call (518) 388-6161 or visit the Wicker Wellness Center in person any weekday between 8:30 and 5:00 to schedule an initial contact appointment. Visit the Counseling Center website for more information.

In a crisis situation, or after hours, contact Campus Safety at (518) 388-6911. The National Suicide Prevention hotline also offers a 24-hour hotline at (800) 273-8255.

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students (dos_office@union.edu) for support or drop into office hours Monday - Friday 8:30 am - 5:00 pm in the Reamer Campus Center room 306. Furthermore, please notify me if you are confortable doing so and I will provide any resources I can. The Union College Persistance Fund can provide financial support in times of unexpected harship to cover needs such as emergency medical expenses not covered by insurance and basic living expenses. Additional sources of supoprt are outlines on the Dean of Students webpage: https://www.union.edu/dean-students .

14.1 ADDENDA

Any community agreed upon additions:

15 TENTATIVE SCHEDULE

- showcase?
- final drafts?

W	TOPIC	Due	Notes
1	Introductions	Onboarding ⍰	mtg 9/10
2		. ⍰	9/17
3		Practice Pitches ⍰	9/24
4		Elevator Pitches ⍰	10/1
5		⍰	10/8
6		⍰	10/15
7		⍰	10/22
8		⍰	10/29
9		⍰	11/5
10			11/12